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CHAPTER 4

PART 3

4.5 CONSTRUCTION

4.5.1 Supervision of Construction

4.5.1.1 The engineer **must supervise the construction** work to ensure that the structure is constructed according to the design and specifications.

4.5.1.2 The responsibilities of the engineer include: -

- 1. Inspection:** All stages of construction are inspected to ensure compliance.
- 2. Approval:** Approval (or disapproval) is given for each construction stage.
- 3. Material testing:** All materials are tested to ensure compliance.
- 4. Site meetings:** The engineer attends and participates in all site meetings.

4.5.2 Parties Involved in Construction

Five parties are involved in construction: -

- 4.5.2.1 Main contractor:** The main construction work is done by the main contractor using: -
- 1. Labour:** Skilled labour (**artisans**) and unskilled labour (**casuals**).
 - 2. Supervisor(s):** Technician(s) in various fields.
 - 3. Site manager:** Usually a holder of Bachelor of Construction Management.
 - 4. Site engineer(s):** Structural, Mechanical and Electrical engineers [in large projects].

4.5.2.2 **Sub-contractors:** Specialist works are done by specialist sub-contractors: -

1. **Electrical sub-contractor:** Responsible for electrical wiring and fittings.
2. **Plumbing sub-contractor:** Responsible for water/waste water piping and fittings.
3. **Air conditioning sub-contractor:**
Responsible for air conditioning equipment.
4. **Network sub-contractor:** Responsible for wiring, equipment and connectivity.

4.5.2.3 **Consultants:** The consultant team consists of : -

1. **Clerk of Works:** Responsible for daily work records. Reports to consultants.
2. **Architect:** Responsible for architectural works, team leader and convener.
3. **Quantity Surveyor:** Responsible for work measurement and certification.
4. **Civil Engineer:** Responsible for civil and structural works.
5. **Electrical Engineer:** Responsible for electrical works and networking.
6. **Mechanical Engineer:** Responsible for plumbing and air conditioning.

4.5.2.4 Client: The client may attend site meetings in person, or may appoint an agent to represent him or her.

4.5.2.5 Regulatory Authorities: Officers from the construction regulatory authorities visit the site regularly to inspect the construction work and ensure compliance with regulations.

4.6 MAINTENANCE

4.6.1 Maintenance Levels

- 4.6.1.1 The term **maintenance** refers to the **activities undertaken** in order to keep the facility **functioning as designed** and to **prevent deterioration** of construction materials.
- 4.6.1.2 The activities depend on the type of facility.
For example, the maintenance of a building is different from the maintenance of a road.

4.6.1.3 Maintenance is done at three levels: -

1. **Routine maintenance** [carried out by the client's in-house staff].
2. **Scheduled maintenance** [carried out by an out-sourced technical team].
3. **Upgrade, facelift or rehabilitation** [carried out by a qualified contractor under the supervision of the design consultants].

4.6.2 Routine Maintenance

- 4.6.2.1 In the case of buildings, routine maintenance includes general cleaning and replacement of burned out bulbs.
- 4.6.2.2 In the case of roads, routine maintenance includes clearing bushes along the road reserve and unblocking drainage channels.

4.6.3 Scheduled Maintenance

- 4.6.3.1 In the case of buildings, scheduled maintenance includes cleaning the outside walls and windows using building maintenance unit, paint work, checks on electrical and mechanical components to assess their working condition and take necessary remedial actions.
- 4.6.3.2 In the case of roads, scheduled maintenance includes repairing road shoulders, repairing pot holes and replacing vandalized road furniture.

4.6.4 Rehabilitation, Facelift or Upgrade

- 4.6.4.1 In the case of buildings, **rehabilitation** or **facelift** includes a complete redesign of the facade, complete change of the architectural finishes, complete change of electrical, plumbing and air conditioning systems.
- 4.6.4.2 In the case of roads, **rehabilitation** or **upgrade** includes replacement of the wearing tarmac layer, widening of the road, change from two-way single carriage to one-way dual carriage, reconstruction of the shoulders and drainage channels.

4.7 SAFE DEMOLITION

4.7.1 Demolition

- 4.7.1.1 **Demolition** is the process of wrecking a structure in such a manner that all the **components and materials go to waste.**
- 4.7.1.2 The **demolition strategy** is to **undermine the stability and strength of the structure** while deliberately controlling the manner and the direction in which the structure will collapse.
- 4.7.1.3 **Safe demolition** requires **special skills.**

4.7.2 Deconstruction

- 4.7.2.1 **Deconstruction** is the process of **dismantling** a structure in such a manner that all the **recoverable components and materials** are **preserved** for possible re-use.
- 4.7.2.2 The **deconstruction strategy** is to **take the reverse route** to that taken during **construction**.
- 4.7.2.3 Deconstruction can be undertaken by the usual skilled construction workers.

4.7.3 Small and Large Structures

- 4.7.3.1 The safe demolition process for “**small structures**” is rather **simple and relatively safe**. The structure is wrecked using construction equipment such as excavators and bulldozers.
- 4.7.3.2 The safe demolition process for “**large structures**” can be risky because of the weight of the structure. If the structure collapses in an uncontrolled manner, it can cause loss of lives or damage to other property or both.

4.7.4 Preparation for Safe Demolition

Before safe demolition starts: -

1. Evacuate all occupants.
2. Disconnect all utilities.
3. Develop safety precautions.
4. Develop demolition work plan.
5. Obtain the required approvals.

4.7.5 Safe Demolition Techniques

Safe demolition techniques include: -

1. High reach excavators
2. Wrecking balls
3. Rotational hydraulic shears
4. Hydraulic hammers
5. Explosives

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THE END